THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mass number of an atom with 6 protons and 6 neutrons is: 1

(a) 18 **(b)** 6 **(c)** 0 **(d)** 12

2. Chadwick discovered the: 1

(a) nucleus **(b)** electron **(c)** neutron **(d)** proton

3. The valency of neon is: 1

(a) 2 **(b)** 0 **(c)** 1 **(d)** 8

4. The number of neutrons in $^{14}_{6}C$ is: 1

(a) 14 **(b)** 6 **(c)** 20 **(d)** 8

5. Thomson discovered the: 1

(a) electron **(b)** nucleus **(c)** neutron **(d)** proton

6. The number of protons in carbon is: 1

(a) 6 **(b)** 4 **(c)** 12 **(d)** 8

7. An atom is electrically: 1

(a) neutral **(b)** unstable **(c)** negative **(d)** positive

8. Bohr proposed: 1

(a) nucleus **(b)** shells (orbits) **(c)** neutron **(d)** plum pudding

9. The number of electrons in a neutral oxygen atom is: 1

(a) 8 **(b)** 16 **(c)** 2 **(d)** 6

10. The proton and the neutron are nearly equal in: 1

(a) size only **(b)** colour **(c)** mass **(d)** charge

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the octet rule? 2

12. What is mass number? 2

13. Write the electronic configuration of nitrogen. 2

14. Who proposed the shell model? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the number of p, e, n in $^{23}_{11}\mathrm{Na}$. 3

16. Write three isotopes of hydrogen with their mass numbers. 3

17. Name the elements with atomic numbers 1, 6, 8. 3

OR

Write the electronic configurations of H, He, Li. 3

18. State three differences between electron, proton and neutron. 3

OR

Name the elements with Z = 11, 12, 17. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Write the electronic configuration of the first 10 elements. 5

OR

Explain the properties of the three sub-atomic particles. 5

20. Explain atomic number and mass number with the example of sodium. 5

OR

Explain the octet rule and stability of noble gases. 5

END OF QUESTION PAPER — Total: 40 Marks

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